STA301 - Statistics and Probability (Quiz 4)

Question # 9 of 10 (Start time: 02:53:18 PM, 07 July 2025)

Critical region is computed with the value of ____

Select the correct option

○	ß
○	1- ß
●	α
○	1-α

STA301 - Statistics and Probability (Quiz 4)

Question # 5 of 10 (Start time: 02:48:25 PM, 07 July 2025)

Critical region is computed with the value of .....

Select the correct option

○	1- β
○	β
○	1-α
◉	α

STA301 - Statistics and Probability (Quiz 4)

Question # 9 of 10 (Start time: 02:43:19 PM, 07 July 2025)

The critical region is computed from the value of:

Select the correct option

○	1-Beta
○	Beta
○	1-alpha
●	Alpha

STA301 - Statistics and Probability (Quiz 4)

Question # 2 of 10 (Start time: 02:44:28 PM, 07 July 2025)

Critical region is computed with the value of .....

Select the correct option

- [x] α
- [ ] $1-\beta$
- [ ] β
- [ ] $1-\alpha$

STA301 - Statistics and Probability (Quiz 4)

Question # 4 of 10 (Start time: 02:48:09 PM, 07 July 2025)

Alpha is the probability of:

Select the correct option

○	Type I error
●	All of these
○	Committing Type I error
○	Reject H0/H0 is true

STA301 - Statistics and Probability (Quiz 4)

Question # 7 of 10 (Start time: 02:48:51 PM, 07 July 2025)

The degrees of freedom associated with paired t-test .....

Select the correct option

○	n1-2 and n2-2
○	n-2
◉	n-1
○	n1+n2-4

STA301 - Statistics and Probability (Quiz 4)

Question # 9 of 10 (Start time: 02:49:20 PM, 07 July 2025)

The level of significance is also called:

Select the correct option

○	Power of test
○	Type II error
○	Acceptence region
●	Type I error

STA301 - Statistics and Probability (Quiz 4)

Question # 3 of 10 (Start time: 02:47:58 PM, 07 July 2025)

By increasing sample size, _____ can be decreased:

Select the correct option

○	Both Alpha and Beta
◉	Beta
○	Alpha
○	None of these

STA301 - Statistics and Probability (Quiz 4)

Question # 1 of 10 (Start time: 02:44:14 PM, 07 July 2025)

Probability of making type II error is reprented by:

Select the correct option

○	1-β
○	α
●	β
○	1-α

STA301 - Statistics and Probability (Quiz 4)

Question # 5 of 10 (Start time: 02:37:59 PM, 07 July 2025)

The "1-alpha" is the probability of:

Select the correct option

○	Type-II error
○	Type-I error
○	Rejection region
◉	Acceptance region

STA301 - Statistics and Probability (Quiz 4)

Question # 9 of 10 (Start time: 02:39:31 PM, 07 July 2025)

For a particular hypothesis test, α=0.05and β=0.05. The power of test is equal to:

Select the correct option

○	0.90
○	0.50
●	0.95
○	0.05

STA301 - Statistics and Probability (Quiz 4)

Question # 6 of 10 (Start time: 02:52:37 PM, 07 July 2025)

In a t-distribution

Select the correct option

○	Median >Mean>Mode
●	Mean=median=mode
○	Mean>Median<Mode
○	Media<Mode<Mean

STA301 - Statistics and Probability (Quiz 4)

Question # 2 of 10 (Start time: 02:37:04 PM, 07 July 2025)

To determine a sample size which test we use .....

Select the correct option

- (●) z
- ( ) t
- ( ) F
- ( ) None of these

STA301 - Statistics and Probability (Quiz 4)

Question # 3 of 10 (Start time: 02:41:32 PM, 07 July 2025)

Interval estimation and Confidence interval are :

Select the correct option

- (●) Same
- ( ) Different
- ( ) Dependent
- ( ) Independent

Question # 8 of 10 (Start time: 02:49:02 PM, 07 July 2025)

To test the hypothesis about the difference of means, which test we use for testing?

Select the correct option

◉	Both a and b
○	F
○	t
○	z

STA301 - Statistics and Probability (Quiz 4)

Question # 4 of 10 (Start time: 02:44:58 PM, 07 July 2025)

t-distribtion is applicable in case of

Select the correct option

◉	Both independent and dependent samples
○	Dependent samples
○	Non-Normal populations
○	Independent samples

STA301 - Statistics and Probability (Quiz 4)

Question # 8 of 10 (Start time: 02:39:17 PM, 07 July 2025)

Which one of the following provides the basis for hypothesis testing?

Select the correct option

○	Critical value
◉	Null hypothesis
○	Test-statistic
○	Alternative hypothesis

STA301 - Statistics and Probability (Quiz 4)

Question # 6 of 10 (Start time: 02:45:28 PM, 07 July 2025)

Interval estimation and Confidence interval are :

Select the correct option

○	Independent
○	Dependent
●	Same
○	Different

STA301 - Statistics and Probability (Quiz 4)

Question # 4 of 10 (Start time: 02:51:57 PM, 07 July 2025)

_____ are equivalent.

Select the correct option

- (●) Type-I error and Level of significance
- ( ) Type-I error and Level of confidence
- ( ) Type-II error and Level of confidence
- ( ) Type-II error and Level of significance

STA301 - Statistics and Probability (Quiz 4)

Question # 8 of 10 (Start time: 02:53:04 PM, 07 July 2025)

The t-distribution can never become narrower than:

Select the correct option

○	exponential distribution
○	F-distribution
●	standard normal distribution
○	Chi-square distribution

STA301 - Statistics and Probability (Quiz 4)

Question # 3 of 10 (Start time: 02:51:39 PM, 07 July 2025)

Conventionally, the probability of making a type-II error is denoted by:

Select the correct option

○	θ (theta)
○	δ (sigma)
○	α (alpha)
●	ß (beta)

STA301 - Statistics and Probability (Quiz 4)

Question # 8 of 10 (Start time: 02:42:57 PM, 07 July 2025)

Alpha is the probability of .....

Select the correct option

●	Making Type I error
○	Making Type II error
○	Rejecting HI when it is wrong
○	Accepting Ho when it is true

STA301 - Statistics and Probability (Quiz 4)

Question # 1 of 10 (Start time: 02:51:11 PM, 07 July 2025)

We can apply t-distribution to test the hypothesis, when

Select the correct option

○	Sample is large
●	Population variance is unknown and Sample is small
○	Sample is small
○	Population variance is known

STA301 - Statistics and Probability (Quiz 4)

Question # 7 of 10 (Start time: 02:45:42 PM, 07 July 2025)

_ _ _ _ _ _ _ _ _ _ _ are equivalent.

Select the correct option

○	Type-I error and Level of confidence
●	Type-I error and Level of significance
○	Type-II error and Level of confidence
○	Type-II error and Level of significance

STA301 - Statistics and Probability (Quiz 4)

Question # 2 of 10 (Start time: 02:51:26 PM, 07 July 2025)

Which one of the following provides the basis for hypothesis testing?

Select the correct option

○	Test-statistic
○	Critical value
●	Null hypothesis
○	Alternative hypothesis

STA301 - Statistics and Probability (Quiz 4)

Question # 10 of 10 (Start time: 02:39:44 PM, 07 July 2025)

If we draw a sample of size 'n' from a population then sample is called "large sample" if:

Select the correct option

○	n>50
○	n>25
●	n>30
○	n>40

STA301 - Statistics and Probability (Quiz 4)

Question # 10 of 10 (Start time: 02:53:31 PM, 07 July 2025)

To test the hypothesis about the difference of means for large samples, what test statistic will be used?

Select the correct option

○	t
○	F
○	Chi-square
●	Z

STA301 - Statistics and Probability (Quiz 4)

Question # 10 of 10 (Start time: 02:43:33 PM, 07 July 2025)

If we are testing H0 : u 1- u 2 = 10 against H1: u 1 - u 2 < 10.Then rejection region will be:

Select the correct option

○	on right side of Z=0
○	on both sides of Z=0
◉	on left side of Z=0
○	in Center

STA301 - Statistics and Probability (Quiz 4)

Question # 5 of 10 (Start time: 02:45:14 PM, 07 July 2025)

The test statistic to test the Ul = U2 (U represent the mean of population)for normal population when n>30.

Select the correct option

○	F-Test
◉	Z-Test
○	All of Above
○	T-Test

STA301 - Statistics and Probability (Quiz 4)

Question # 10 of 10 (Start time: 02:49:33 PM, 07 July 2025)

For test of hypothesis Ho:μ 1 = μ 2 and H1:μ1 < μ2, the critical region at 0.05 level of significance and n > 30

Select the correct option

○	Z>1.645
◉	Z<-1.645
○	Z less and equal 1.96
○	Z> 1.96

STA301 - Statistics and Probability (Quiz 4)

Question # 4 of 10 (Start time: 02:41:47 PM, 07 July 2025)

The proportion of working females in Pakistan is at most 0.40,the null hypothesis Ho<0.40 the alternative hypothesis Hl is:

Select the correct option

◉	P > 0.40
○	P = 0.40
○	P < 0.40
○	P ≠ 0.40

STA301 - Statistics and Probability (Quiz 4)

Question # 2 of 10 (Start time: 02:47:44 PM, 07 July 2025)

To test the hypothesis about the difference of means for small samples, test statistics will be used:

Select the correct option

○	F-test
◉	T-test
○	All of these
○	Z-test

STA301 - Statistics and Probability (Quiz 4)

Question # 5 of 10 (Start time: 02:42:07 PM, 07 July 2025)

The t-distribution is applicable for testing about the population mean, when population standard deviation is:

Select the correct option

○	Greater then 40
○	Known
◉	Unknown
○	Less than 40

STA301 - Statistics and Probability (Quiz 4)

Question # 7 of 10 (Start time: 02:39:00 PM, 07 July 2025)

"The temperatures of patients were checked before and after given a certain medicine."
This is an example of:

Select the correct option

○	Natural pairing of data
○	Automatic grouping
◉	Pairing by design
○	Independent data

STA301 - Statistics and Probability (Quiz 4)

Question # 6 of 10 (Start time: 02:38:42 PM, 07 July 2025)

For the smaller values of ν, the t-distribution will be _____ than the standard normal distribution.

Select the correct option

○	None of these
●	more spread out
○	same length
○	less spread out

STA301 - Statistics and Probability (Quiz 4)

Question # 3 of 10 (Start time: 02:44:42 PM, 07 July 2025)

The proportion of males in Pakistan is at least 0.48, the Null Hypothesis is $p \geq 0.48$, then the alternative hypothesis Hl is

Select the correct option

○	P = 0.48
○	$P \geq 0.48$
○	$P \leq 0.48$
◉	P < 0.48

STA301 - Statistics and Probability (Quiz 4)

Question # 10 of 10 (Start time: 02:46:34 PM, 07 July 2025)

The proportion of working females in Pakistan is at most 0.40,the null hypothesis Ho<0.40 the alternative hypothesis Hl is:

Select the correct option

◉	P > 0.40
○	$P \pm 0.40$
○	P = 0.40
○	P < 0.40

STA301 - Statistics and Probability (Quiz 4)

Question # 4 of 10 (Start time: 02:37:37 PM, 07 July 2025)

If we are testing H0 : u 1- u 2 = 10 against H1: u 1 - u 2 < 10.Then rejection region will be:

Select the correct option

○	on right side of Z=0
○	on both sides of Z=0
○	in Center
◉	on left side of Z=0

STA301 - Statistics and Probability (Quiz 4)

Question # 2 of 10 (Start time: 02:41:16 PM, 07 July 2025)

The proportion of males in Pakistan is at least 0.48, the Null Hypothesis is $p \geq 0.48$, then the alternative hypothesis H1 is

Select the correct option

○	$P \leq 0.48$
○	$P = 0.48$
●	$P < 0.48$
○	$P \geq 0.48$

STA301 - Statistics and Probability (Quiz 4)

Question # 3 of 10 (Start time: 02:37:24 PM, 07 July 2025)

For interpreting the value of t using a t-distribution table, the factors that must be known are:

Select the correct option

○	The significance level
○	Whether hypothesis is one or two-tailed
●	All of the above
○	The degrees of freedom

STA301 - Statistics and Probability (Quiz 4)

Question # 9 of 10 (Start time: 02:46:19 PM, 07 July 2025)

For the smaller values of v, the t-distribution will be _____ than the standard normal distribution.

Select the correct option

- (●) more spread out
- ( ) same length
- ( ) less spread out
- ( ) None of these

STA301 - Statistics and Probability (Quiz 4)

Question # 1 of 10 (Start time: 02:36:50 PM, 07 July 2025)

In hypothesis testing, suppose the critical value is |Z| > 2.33. This value shows that the applied test is .....?

Select the correct option

○	Right tailed test
○	Left tailed test
○	One tailed test
◉	Two tailed test

STA301 - Statistics and Probability (Quiz 4)

Question # 6 of 10 (Start time: 02:42:22 PM, 07 July 2025)

Suppose H0: p =0 H1:p < 0 and Critical value is z = - 2.33. If calculate the value of Z = - 2.60,then what will be your conclusion?

Select the correct option

○	None of the above
○	Impossible to decide
○	Accept Ho
◉	Reject Ho

STA301 - Statistics and Probability (Quiz 4)

Question # 7 of 10 (Start time: 02:42:38 PM, 07 July 2025)

The proportion of working females in Pakistan is at most 0.40, the Null Hypothesis is H0 : P≤ 0.40. then the alternative hypothesis H1 is

Select the correct option

- ○ none of these
- ○ P < 0.40
- ○ P = 0.40
- ◉ P > 0.40

STA301 - Statistics and Probability (Quiz 4)

Question # 8 of 10 (Start time: 02:46:04 PM, 07 July 2025)

Suppose H0 : u 1 = u2 and H1:u1 >u 2 and Critical value is z= 1.645. If calculated value of Z = 1.32,then what will be your conclusion?

Select the correct option

- ○ Reject Ho
- ○ Impossible to decide
- ○ None of the above
- ◉ Accept Ho

STA301 - Statistics and Probability (Quiz 4)

Question # 1 of 10 (Start time: 02:47:31 PM, 07 July 2025)

The proportion of working females in Pakistan is at most 0.40, the Null Hypothesis is H0 : P≤ 0.40. then the alternative hypothesis H1 is

Select the correct option

- P < 0.40
- P = 0.40
- none of these
- P > 0.40 (selected)

STA301 - Statistics and Probability (Quiz 4)

Question # 6 of 10 (Start time: 02:48:39 PM, 07 July 2025)

Suppose H0 : u 1 = u2 and H1:u1 >u 2 and Critical value is z= 1.645. If calculated value of Z = 1.32,then what will be your conclusion?

Select the correct option

- Reject Ho
- None of the above
- Impossible to decide
- Accept Ho (selected)

STA301 - Statistics and Probability (Quiz 4) Quiz Start Time: 02:51 PM, 07 July 2025

Question # 5 of 10 (Start time: 02:52:22 PM, 07 July 2025) Total Marks: 1

When the population is normally distributed, population standard deviation is unknown and the sample size is n = 15, the confidence interval for the population mean is based on

Select the correct option

- the z (standard normal)distribution.
- (●) the t distribution.
- the Poisson Distribution.
- the Binomial distribution.

Click to Save Answer & Move to Next Question

STA301 - Statistics and Probability (Quiz 4) Quiz Start Time: 02:40 PM, 07 July 2025

Question # 1 of 10 (Start time: 02:40:45 PM, 07 July 2025) Total Marks: 1

An experiment was conducted to estimate the mean yield of a new variety of wheat. A sample of 20 plots gave a mean yield of 2.9, and a 95% confidence interval of (2.48, 3.32) . This means:

Select the correct option

- None of these
- We are sure the true mean yield of this new variety is between 2.48 and 3.32.
- (●) We are 95% con?dent that the true mean yield of this variety is between 2.48 and 3.32
- We are 95% confident that the true mean yield of this variety is 2.9

Question # 7 of 10 (Start time: 02:52:50 PM, 07 July 2025) **Total Marks:** 1

An experiment was conducted to estimate the mean yield of a new variety of wheat. A sample of 20 plots gave a mean yield of 2.9, and a 95% confidence interval of (2.48, 3.32) . This means:

Select the correct option

- ○ We are 95% confident that the true mean yield of this variety is 2.9
- ○ We are sure the true mean yield of this new variety is between 2.48 and 3.32.
- ◉ We are 95% con?dent that the true mean yield of this variety is between 2.48 and 3.32
- ○ None of these

Click to Save Answer & Move to Next Question